Data Analytics Model to Reduce Rejection Rates in Small Industries

Abstract

This project presents a data-driven analytics and machine learning system designed to reduce product rejection rates in small-scale manufacturing industries. The system collects production data, performs statistical analysis, identifies root causes of defects, and predicts defective products using machine learning models.

Problem Statement

Small industries face high rejection rates due to lack of structured data analysis, manual inspection processes, and absence of predictive systems. This results in financial losses and reduced productivity.

Objectives

1. Collect structured production data.
2. Analyze defect trends and patterns.
3. Identify root causes.
4. Predict defective products using ML.
5. Reduce rejection rates and improve profitability.

System Architecture

Production Data → CSV/Database → Python Analytics → Machine Learning Model → Dashboard → Actionable Insights.

Dataset Structure

Machine_ID, Operator_ID, Shift, Temperature, Pressure, Material_Batch, Rejected.
Rejected = 1 (Defective), 0 (Good Product).

Methodology

1. Data Collection
2. Data Cleaning & Preprocessing
3. Exploratory Data Analysis
4. Model Training (Random Forest)
5. Dashboard Visualization
6. Root Cause Identification

Machine Learning Model

Random Forest Classifier is used for prediction. The model is trained using historical production data and achieves high accuracy in identifying potential defective products.

Results

The system identifies high-risk conditions such as high temperature, excessive pressure, and night shift production. Expected rejection rate reduction: 20–40%.

Cost Benefit Analysis

Example: If 10,000 units are produced monthly with 12% rejection and ■50 loss per unit:
Monthly Loss = ■60,000
Reducing rejection to 6% saves ■30,000/month
Annual Savings = ■3,60,000

Conclusion

This project demonstrates how data analytics and machine learning can significantly improve quality control in small industries. The system enables data-driven decision making and reduces operational losses.

Future Scope

1. Real-time IoT integration
2. AI-based defect image detection
3. Predictive maintenance
4. Cloud deployment
5. Automated report generation